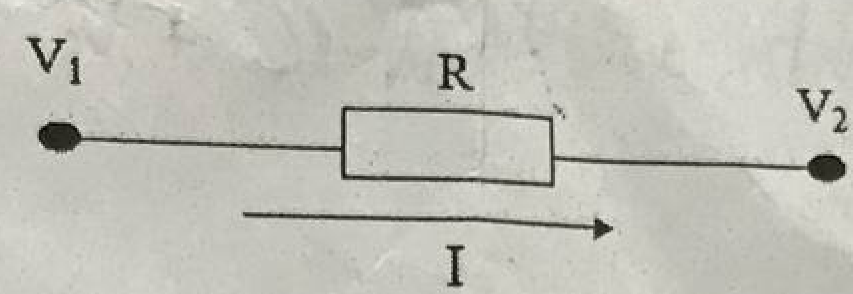
National University of Computer and Emerging Sciences, Lahore Campus

Course:	Instrumentation and Measurement	Course Code:	EE220
Program:	BS(Electrical Engineering)	Semester:	Fall 2018
Duration:	60 Minutes	Total Marks:	30
Paper Date:	03-Oct-2018	Weight	15%
Section:	ALL	Page(s):	1
Exam:	Midterm-I	Questions	3

Q1 The voltage (V_1) at one ends of 500 $\Omega \pm 5\%$ resistor (R) is measured as $V_1 = 10 \pm 0.5$ V. The current through the resistor is $I = 10\text{mA} \pm 5\%$. Find the magnitude and relative accuracy of the voltage V_2. [10] CLO 01



25.5%

Q2 A rectifier ammeter is to indicate full scale for a 1 V RMS primary current. The PMMC instrument used has 500 μA FSD and the coil resistance $R_M = 1200\ \Omega$. The current transformer turns are $N_S = 7000$ and $N_P = 10$. Silicon diodes are used and the meter series resistance is $R_S = 150$ kΩ. [10] CLO 02

a. Draw the circuit diagram.
b. Determine the required secondary shunt resistance (R_L) value.

Q3 A series ohmeter indicates 80% ***ohmeter deflection*** when measuring a resistance of R_x 100 kΩ. The PMMC instrument used has 1.0 mA FSD and the coil resistance R_M = 500 Ω. [10] CLO 02

a. Draw the circuit diagram.
b. Determine the values of series-resistance (R_S) and the battery-voltage (V_B).

$20.1 = 0.2(500 \times$

0.2

0.2mA

$V = IR_m + IR_S + IR_x$.

$= I(R_m + R_S + R_x)$

$\frac{N_S}{N_P} = \frac{V_S}{V_P}$

$\frac{N_S}{N_P} = \frac{I_P}{I_S}$

$\frac{N_P}{N_S} \times I_P = I_S$

$= 700$ V: